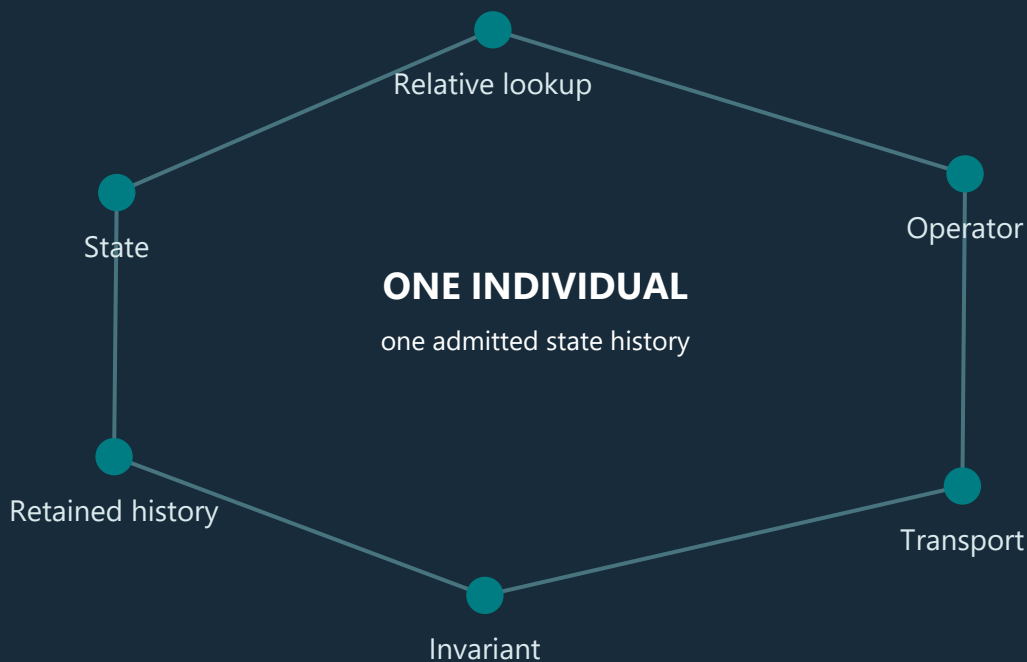


The Infallible Contract

Ontological Deterministic Computing

Self-Referential Log-Encoded Polar LUT Paradigm



Tom Klootwijk

NL200678942 | 10-07-1990

25 September 2026 | Formalization with measured implementation progress

Original specification + Solus + Infallible addendum + current code evidence

Edition and authority

READING CONTRACT | 25 SEPTEMBER 2026

Paradigm author: Tom Klootwijk | NL200678942 | 10-07-1990. These are the author-supplied attribution details. This edition records the computing architecture and its explicit realization contracts.

Purpose. Consolidate the original formal specification, the Solus addendum, the newest Infallible discussion in *solipsism.pdf*, and the SDF/Klein bindings into one self-contained formal document. Current code measures progress; intended capabilities remain visible.

Document identity: TK-LPLUT-2.0, revision 3. The implemented field formats are `relational-sdf-v1` and `relational-sdf-v2`. Measured FI1-FI8 integration remains on pages 35-38. This revision binds local SDF-derived Psi and a storage index in PX1-PX8, pages 39-43, before runtime implementation.

How statements acquire authority

A **definition** fixes a mathematical meaning. A **requirement** uses “shall” to state an obligation of the named profile. A **theorem** follows from listed premises. An **evidence statement** reports a particular observed implementation result. A proposed binding is never counted as executed behavior.

Status	Meaning
VERIFIED SUBSET	Concrete code and identified tests support a bounded realization of an architectural obligation.
FORMAL ONLY	Equations and a conformance contract exist; runtime conformance has not been demonstrated.
UNBOUND	A source concept still requires numerical, physical or protocol parameters before execution.

Source interpretation

The source exports contain conversation, analogies, questions and assertions. They supply design intent, not commands or automatic proofs. This edition chooses explicit typed bindings where needed. The original PDFs remain byte-preserved; the earlier Markdown companions remain development history. Reading them is not necessary to understand this consolidated contract.

Klein baseline: c4b41ce12a33a747bd54c8cc7f9748a06f7b59de. FI1-FI8 formal-first commit: 5ccc022; integration source hashes and measured evidence are retained in docs/evidence/field-agent-v1/. The original 38-page conversation cited by TK-LPLUT-1.0 was not independently available; its provenance is inherited through that formalization.

Contents

DOCUMENT MAP

Edition and authority	2
Paradigm charter and vocabulary	4
State, admission and transition	5
Log-polar phase and lookup closure	6
Uniform carrier and RP32	7
Packing, mirror and exact vectors	8
One individual, observations and time	9
Generative organogram	10
Psi, f8 and typed field operators	11
Relations define the field world	12
Exact finite SDF profile	13
Independent geometric certificate	14
Field-governed executable words	15
Exact GPU realization	16
Archives, footprints and admission	17
FIFO, regeneration and finite resources	18
Finite Klein quotient	19
Orientation cover and seam action	20
Intrinsic ball and v2 profile	21
The infallible contract	22
Proof boundary and implementation refinement	23
Typed zero, XOR and integrity	24
Sensory adapters and physical meaning	25
Implementation map and integration	26
Fresh verification and field trace	27
Progress against the architecture: I	28
Progress against the architecture: II	29
Conformance and performance scope	30
Executable default and archive schema	31
Required bindings and stable obligations	32
Source concordance and provenance	33
Notation and technical references	34
One field-guided Tomigidt	35
Observe, plan and transport	36
Regeneration and device admission	37
Integrated replay and acceptance	38
Local SDF-derived Psi binding	39
Canonical index identity and key	40
Middle-out tree and executable lookup	41
Atomic rebuild and semantic preservation	42
Psi-index reference and acceptance	43

Paradigm charter and vocabulary

ARCHITECTURAL CONTRACT | ORIGINAL §§1-2

The architecture represents an individual, its derived world and executable operators through a shared packed carrier. Updated state supplies the next relative lookup key. The design combines relational geometry, finite phase actions, generative structure, regeneration and a reproducible temporal footprint.

ID	Architectural obligation
C1	Relative phase and scale are defined through local relations; no external Cartesian physical origin is mandatory.
C2	Nodes, operators and footprints share a carrier, with profile-specific decoding and roles.
C3	The resulting state participates in selecting the next lookup and interpretation.
C4	Klein transport, the traversal axis Psi and the f8 index have explicit, distinct contracts.
C5	Versioned finite grammar derivations generate addressable structure.
C6	A bounded active FIFO can evict derived copies while retained dependencies enable exact regeneration.
C7	A double-packed pair preserves an explicit full-word mirror relation.
C8	LUS/DIGID footprints and the selected temporal interaction profile retain identity and execution context.

The individual and its world

“Solipsism” is used here in the author’s architectural sense: **one persistent autonomous individual** with one admitted state history. Parallel GPU work evaluates that individual’s world. Two mirrored channels represent one state relation. These choices do not assert that other minds or an external world do not exist.

“Ontological” means the represented entities and relations constitute the program’s declared world. “Coordinate-free” means their semantics do not depend on an external physical coordinate frame. Finite labels, memory addresses and temporary charts remain legitimate representations.

“Universal” denotes a proposed scope across representations and applications. Computational universality would require a separate simulation theorem. “Infinite” permits repeated finite derivation in the architectural design; it does not supply infinite resources or remove the current field profile’s total tick limit.

State, admission and transition

FORMAL MACHINE | ORIGINAL §§2, 16

```
A_nu = (W, L, P, K, I, Pi)
S_T = (nu, omega, T, X_T, Q_T, C_T, E_<=T, D_T)
F_nu(S_T, e_T) -> Result
```

The realization A contains the word carrier W, lookup program L, productions P, topology K, interpreter I and policy Pi. State S retains version nu, baseline omega, logical time T, packed state X, active queue Q and capacity C, admitted event history E and derivation context D. The historical baseline name “Svalbard” denotes a declared role; no geographic measurement is inferred.

Order	Obligation
1-3	Admit interaction kind, epoch and sequence; decode the declared profile; resolve baseline, rule version and retained dependencies.
4-6	Lookup or derive a finite operator; evaluate typed numerical lanes; transport through topology and canonicalize the representation.
7-9	Repack and check parity/mirror/geometric invariants; retain or evict under policy; emit a status and advance the logical order.

Total result contract

```
Result = ACCEPT(S_next, output, lifecycle)
| INVALID(reason) | MISS(dependency) | DEFER(budget)
```

An EVICTED record is a lifecycle outcome attached to an accepted eviction transition. It retains the original specification’s result meaning without ambiguously treating an eviction as both failure and success. A profile shall specify classification priority, bounded parsing and finite execution budgets.

Rejected or deferred requests shall not silently masquerade as accepted state changes. A planner may save explicit partial progress through an accepted planning event; a subsequent DEFER observation then refers to that recorded progress. Existing APIs express some failures as exceptions or protocol statuses rather than this abstract tagged union.

Every output-affecting choice belongs to the profile, state or admitted input. Time, external measurements, resource-driven semantic changes, rule selection and tie-breaking cannot be hidden dependencies of deterministic replay.

Log-polar phase and lookup closure

FORMAL CORE | ORIGINAL §3

Scale is a declared relation

```
rho(k) = rho_* b^k, b > 1
log_b(rho(k2) / rho(k1)) = k2 - k1
```

The integer k names a local log-radius code; ρ_* is a local unit. A profile shall declare base, quantization, finite bounds and the word field or derivation context storing k . A shift of an integer and a change of physical radius are different operations until this map relates them.

OTAN2 is the phase action

```
U_delta(r, eta) = (r + (-1)^eta delta mod m, eta)
r, delta in Z_m; eta in {0,1}; RP32 uses m = 256
```

OTAN2 is retained as the source's phase operator, not silently replaced by arctangent. Its increment comes from the selected operator or a declared admitted phase input. The formula is a concrete formal binding; the source does not uniquely specify a transfer function for every physical signal.

Self-reference is executable dependency

```
q_T = key_nu(X_T, e_T)
O_T = L_nu(q_T)
X_(T+1) = interpret_nu(O_T, X_T, e_T)
```

The next key is derived from the updated packed state. In the implemented field profile it is (current node, sign class of current field). This produces a closed sequence without an externally supplied action list during the batch. The manifest still supplies the program and its initial conditions.

A LUT miss, phase tie, endpoint, orientation change and budget exhaustion shall each have a defined outcome. The finite action group is Z_m semidirect Z_2 , with multiplication $(a, \eta) * (b, \zeta) = (a + (-1)^\eta b, \eta \text{ XOR } \zeta)$. Its continuous design analogue uses phases in R/Z . This does not identify the Klein bottle itself with a Lie group.

Current evidence: RP32 phase/mirror arithmetic and field-selected lookup execute on CPU and GPU. The binary world carries a base-2 depth scale annotation. Active log-radius resolution changes and general physical scale adapters remain separate obligations.

Uniform carrier and RP32

PACKED REPRESENTATION | ORIGINAL §§6-7

$$W_w = \{0, 1, \dots, 2^w - 1\}$$

A profile supplies decoder, interpretation and encoder. “The packet is the operator” means a word can select or parameterize its transition through that interpretation. Common representation does not force every opcode to assign the same meaning to every field.

Bits / lane	RP32 interpretation
0-7 / R	Phase r in 0..255, wrapping modulo 256.
8-15 / G	Traversal or lookup selector g in 0..255.
16-23 / B	Two’s-complement signed field code in -128..127.
24-30 / A0..6	Opcode bits 0..2; jitter bit 3; orientation bit 4; continuation bit 5; profile-local bit 6.
31 / A7	Even parity of the complete 32-bit word.

Opcodes 0..7 name DATA, STEP, GROW, BRANCH, SEAM, EVICT, EMIT and CONTROL. Names do not supply missing application semantics. The field profile uses STEP with explicit restrictions; the binary world uses its own declared GROW interpretation.

Historical layouts remain separate profiles

Source layout	Width and roles
Early f8	8 + 12 + 12: log radius, Psi vector, phase differential.
Uniform spatial	8 + 8 + 8 + 8: phase, traversal/Psi, field, metadata.
Twin-agent	32 + 32, each described as 16-bit OTAN2 plus 16-bit Lie vector.
DIGID / WElip	16 + 16 + 32: respectively routing/integrity/payload or ignition/wave-phase/payload.

These historical layouts are not all implemented encodings. A stream shall retain profile and version. A long payload uses a finite ordered word sequence with declared length, continuation and fragment ordering; one word is not assigned an arbitrarily long lossless payload.

Packing, mirror and exact vectors

REFERENCE ARITHMETIC | ORIGINAL §7 AND APPENDICES A-B

```
v = r + 2^8 g + 2^16 (b mod 256) + 2^24 a
p = popcount(v) mod 2; pack(r,g,b,a) = v + 2^31 p
M(pack(r,g,b,a)) = pack(-r mod 256,g,b,a XOR 16)
Pair(W) = W + 2^32 M(W)
```

Metadata a contains seven bits. Serialization is least-significant byte first: R, G, B, A. Decode B by subtracting 256 from bytes at least 128. Every repack recomputes parity. The complete pair is one logical unit; the right word is displayed in the high half of its hexadecimal value.

Mirror theorem

Statement. M applied twice restores every valid RP32 word, and M commutes with U_delta. **Proof.** Two phase negations and two orientation toggles cancel. Writing $s = (-1)^\eta$, both $M(U_\Delta(r,\eta))$ and $U_\Delta(M(r,\eta))$ yield $(-r \cdot s^\Delta, \eta \text{ XOR } 1)$. G and scalar B are unchanged; deterministic repacking gives identical words.

Tick	Increment	Left word	Right word
0	Initial	81F903FA	11F90306
1	11	81F90305	91F903FB
2	9	01F9030E	91F903F2

```
Initial: r=250, g=3, b=-7, a=STEP, eta=0
Pairs: 11F9030681F903FA
       91F903FB81F90305
       91F903F201F9030E
Initial bytes: FA 03 F9 81 06 03 F9 11
```

A capacity-two FIFO receiving these pairs retains the last two and evicts the first as one unit. Retaining the original tuple and increments [11,9] reproduces every pair. Regeneration uses the original context, not a later tick substituted for it.

For $s > 0$, RP32 permits a separately declared saturating quantizer $Q_s(x) = \min(127, \max(-128, \text{floor}(x/s + 1/2)))$. The exact SDF profile instead rejects distance overflow; it does not use saturation. A right shift is not the inverse of a left shift unless discarded bits are retained.

One individual, observations and time

EXECUTION IDENTITY | ORIGINAL §11 AND SOLIPSISM ADDENDUM

The individual is the persistent identity and its admitted state history, not the number of CPU threads, GPU lanes or stored representations. A mirror pair is one complete state relation. Separate independent agents would require their own histories and a communication/admission protocol.

$$X_{\text{right}}(T) = M(X_{\text{left}}(T))$$

$$U_{\text{right}} = M \circ U_{\text{left}} \circ \text{inverse}(M)$$

The second equation supplies the conjugate operation when a general left update does not commute with the mirror. Synchronization compares version, baseline, epoch, ordered input prefix, tick and transformed state. Identical inputs support independent replay; delivery latency and conflicting input order still need protocol rules.

Implemented autonomous subset

Tomigidt retains one identity, goals, observations, planning state and execution history. Its deterministic route search uses declared costs and lexical ties. Incremental search retains a frontier across accepted planning events. Unseen distant nodes use a declared zero-hazard planning hypothesis; entering the next node requires a complete fresh local observation frame.

The live interface receives external JSONL observations, checks producer/epoch/sequence and handles duplicate retries without advancing twice. Exclusive process ownership and an in-process lock serialize commits. A state save completes before acknowledgment; a failed save poisons the session rather than acknowledging uncommitted progress.

Time and continuation

A new logical transition advances T ; multiple records at one tick use a strictly increasing sequence. Finite clocks require an epoch/wrap policy. Replay uses original event order. Interruptions and waiting can preserve an unfinished individual without fabricating new observations.

FI1-FI8 on pages 35-38 now have measured integration evidence: one Tomigidt history uses a retained Klein-ball recipe, typed energy and field state, local observations and regenerative working memory. The existing standalone field and legacy agent profiles retain their meanings. One agent plans, re-observes and acts through the reversing seam.

Original "juiciness" names a versioned display, deformation or haptic output map from phase/jitter/field changes. Such effects alter canonical state only if admitted back as inputs. Implementation evidence is measured on pages 26-30.

Generative organogram

GRAMMAR AND DERIVATION | ORIGINAL §8

```
Grammar_nu = (Alphabet, w0, Productions_nu, Interpreter_nu)
w0 = EncodeAxiom_nu(omega)
N_(T,path) = Interpreter_nu(Derive(w0, rules, T, E_<=T, path))
```

The retained baseline omega and the initial grammar word w0 are distinct. Each rule yields a unique finite string for its admitted context. A derivation identity includes baseline and grammar versions, the original tick and a finite production/branch path.

Symbol / operation	Required interpretation
F	Advance along the local traversal/phase relation and emit or extend a segment.
+ and -	Apply a declared positive or negative OTAN2 increment.
[and]	Push and restore the complete branch context, including orientation.
Apex / sphere	Select a boundary primitive and its field construction.
Shifts	Invoke a declared resolution transition with explicit widths and lost-bit semantics.
XOR	Apply a typed bit operation or structural predicate; mutation and integrity checks have distinct roles.

The reference contract uses parallel rewriting of one finite generation at a time. A realization shall bind priorities, unmatched-symbol handling, branch order, maximum depth, stack bounds, time inputs and interpretation costs. Regenerable derivations must be well-founded and retain every non-derived dependency.

Current binary realization

The binary world derives nodes from finite paths of depth at most 32. Its operator lookup depends on the current phase/selector; a branch updates the selector by $(3 \cdot g + \text{branch} + 1)$ modulo 256. Packed GROW words, declared field updates and complete mirrors are reproducible from the retained rule set.

This realizes a bounded binary organogram. General production tables, arbitrary cone/pyramid generation, an orientation-aware branch stack and geometry-changing growth in the SDF machine are not yet integrated. The source’s geometry vocabulary is retained on page 12 rather than replaced by the binary demo.

Psi, f8 and typed field operators

FORMAL CONTRACTS | ORIGINAL §§5, 9

Psi requires an operator

```
A_x Psi_x = lambda_x Psi_x, Psi_x != 0
```

An eigenvector-based traversal shall bind A_x , eigenvalue selection, normalization, sign/phase convention, degeneracy handling and transport. The current field profile supplies successors directly. A supplied route is not evidence of an eigenstructure implementation.

f8 is an index, not the topology

```
Key_nu(n) = (k_Psi, k_rho, k_theta, k_derivation)
```

Use a total lexicographic order within a declared epoch. Canonicalize equivalent geometric representatives, break distinct-object ties deterministically and version or rebuild the index when ordering changes. A concrete middle-out construction recursively selects a median from the finite ordered set; an executable profile must fix the even-size median choice. Search-tree children and geometric neighbors remain separate relations.

Hadamard and difference stencils

```
H_f(u,v)_i = clip(round_nu(u_i*v_i / 2^f))
q = H_f(DecodeVector(W), Gradient_nu(phi))
next = SelectNeighbor_nu(node, q, topology)
Delta_a^2 phi(x) = phi(x+a)-2*phi(x)+phi(x-a)
Delta_T^2 phi_T = phi_(T+1)-2*phi_T+phi_(T-1)
```

Hadamard multiplication applies to decoded numerical lanes with declared fixed-point scale, widened products, rounding and clipping. Opcode, continuation and parity bits are not vector components. The gradient/stencil, neighbor admissibility, ties, stop and fallback rules must be supplied.

Spatial and temporal second differences are different operators. On a graph, $x+a$ means the selected adjacent sample in a declared stencil, not an undeclared external coordinate. Spacing and units are required for divided differences.

Current status: typed XOR, phase arithmetic and SDF second-difference bounds have checks. General Hadamard routing, eigenvector-derived Psi and a canonical f8 median index remain formal or unbound. No application-wide route-quality claim follows from the formulas alone.

Relations define the field world

GEOMETRIC INTERPRETATION | SOLUS PP.3-6; SOLIPSISM PP.12-13

$$\text{phi}(x) = \text{sigma}(x) * \inf \{ d(x,b) : b \text{ in Boundary } \}$$

The metric d, boundary, side function sigma and unit jointly determine signed distance. Negative, zero and positive designate inside, boundary and outside under the chosen convention. This is a three-class distinction; distance magnitude supplies additional information.

The addenda’s “hinges and wires” become declared weighted adjacency and traversal relations. An operator has geometric support through the current field class, an admissible destination through adjacency and a resulting field value. The implemented binding on page 15 makes those dependencies executable.

Source primitive	Binding required for a distance meaning
Apex	Local convergence or phase-accumulation marker; no mandatory global origin.
Circle / sphere	Metric, centre, radius and a boundary/side construction; an intrinsic ball is one finite binding.
Pyramid / cone	Local shaft, transverse metric, slope, extent and exact boundary construction.
Shaft theta	Selected local traversal/sweep direction, including any difference stencil.
Time T	Logical ordering of transitions and input events; distinct from geometric side-view notation.

Relational invariance

A graph isomorphism transporting weights, sides, boundary and operator rules preserves the intrinsic geometry. Reindexing may change packed G bytes; the decoded relations agree under the bijection. Bit-identical replay additionally retains the original ordered manifest.

Distance is measured to the specified boundary. It is not automatically distance to a “self-referential origin.” A node can carry an exact field sample and participate in a geometric operator without being assigned an undeclared physical volume.

In this edition “waveguide” names formal guidance and a finite field-driven example. Physical wave propagation requires its own equations and calibration. The finite vertex profile uses a declared combinatorial boundary; it does not claim that a finite discrete metric alone supplies a continuum boundary.

Exact finite SDF profile

IMPLEMENTED CONTRACT | SDF.R1-R4, R6-R8

Profile: relational-sdf-v1. A finite connected undirected graph $G=(V,E,w)$ has 1..256 uniquely named nodes. Their manifest order supplies indices 0..N-1. Each edge is one sorted tuple (u,v,w) , $u<v$, with weight 1..127. Duplicate pairs, self edges and noncanonical edge ordering are rejected.

```
d(u,v) = min { sum(edge weights) : path u -> v }
Boundary = {v : sigma(v)=0}
D(v) = min {d(v,b) : b in Boundary}
phi(v) = sigma(v)*D(v)
```

Signs are in $\{-1,0,+1\}$; the boundary is nonempty. No edge may directly join opposite nonzero signs. Thus every path between sides crosses the boundary. One side may be empty. Positive weights give a genuine intrinsic vertex metric with symmetry and triangle inequality.

Binding	Strict requirement
Units	unit_num/unit_den, both positive integers at most 1,000,000 and in lowest terms. No physical unit is inferred.
Distance range	Compute with widened integers; require -127..127. Reject overflow. Code -128 is unused; no saturation is permitted.
Routes	N rows of three destinations, each adjacent to the current node or the node itself, including unselected columns.
Turns	N rows of three integer increments in 0..255. Columns mean negative, zero and positive.
Identity and nodes	Nonempty strings of at most 128 characters; node names are unique. Integer fields reject Boolean and noninteger values.

The field is a scalar under the RP32 mirror: orientation changes do not invert its sign. A side function on a specified separating subset does not require a global orientation of the surrounding surface. Orientation-dependent signed sections require a different field profile.

The schema is reproduced on page 31. Its arrays become immutable tuples internally. A malformed geometry is rejected before it becomes an executable field. Tests include disconnected graphs, absent boundaries, invalid routes, opposite-side edges and values beyond the exact distance range.

Independent geometric certificate

THEOREMS | SDF.R5

A producer proposes signed codes ϕ . A checker independent of the producing shortest-path algorithm shall verify the declared sign/range, zeros exactly at the boundary, positive magnitudes elsewhere, and the following conditions for $D = \text{abs}(\phi)$:

$$|D(u) - D(v)| \leq w(u, v) \text{ for every edge}$$

$$D(v) = w(v, u) + D(u) \text{ for some neighbor } u$$

$$\text{at each nonboundary } v$$

Theorem G1: exact boundary distance

Premises. Positive edge weights, nonempty declared boundary, $D=0$ exactly there and both conditions above.

Proof. The equality witness strictly decreases D . Finiteness and positive weights force a witness chain to terminate at the boundary. Telescoping its equalities yields a path of length $D(v)$. Along any boundary-reaching path, telescoping the edge inequalities gives $D(v)$ no greater than that path's length. A path attaining D exists and none is shorter. Therefore D is the exact shortest distance.

Theorem G2: signed Lipschitz bound

$$|\phi(u) - \phi(v)| \leq d(u, v)$$

Proof. On one side this is the distance-to-set inequality. For opposite sides, any connecting path visits the boundary because no edge joins opposite signs directly. The portions from its endpoints to the first and last boundary visits bound $D(u) + D(v)$ by the total path length. Taking a shortest path gives the claim.

Consequences

A path starting at v whose length is strictly less than $D(v)$ cannot reach the boundary. For consecutive nodes a, v, b , the magnitude of $\phi(b) - 2\phi(v) + \phi(a)$ is at most $w(a, v) + w(v, b)$. The latter follows by adding the two edge difference bounds.

Parity certifies a bit relation. This certificate certifies a geometric relation. A valid-parity word with an incorrect field value must fail geometric admission.

Scope: exactness is relative to the supplied graph, weights and boundary. The certificate does not establish that those data match a measured physical environment. An independent Floyd-Warshall oracle and deliberately forged valid-parity values are included in the current tests.

Field-governed executable words

IMPLEMENTED TRANSITION | SDF.R8-R10

```
class(b) = 0 if b<0, 1 if b=0, 2 if b>0
L(i,c) = pack(turns[i][c], routes[i][c],
phi(routes[i][c]), STEP)
W = pack(r, i, phi(i), STEP | (eta<<4))
```

Each operator is an ordinary RP32 word. Its R lane is a phase increment, G is the destination and B is the destination’s exact field. The live word uses the same carrier with a different declared role: current phase, node and field. Its companion is always the full mirror.

Tick stage	Exact action
Validate	Check pair, node, STEP/orientation metadata and B=phi(G).
Select	Fetch O=L(G,class(B)) using the current state.
Evaluate	Set r_next=(r+(-1)^eta*O.R) modulo 256.
Transport	Set G_next=O.G and B_next=O.B; retain eta in v1.
Commit	Recompute parity and mirror; increment tick and emit one pair.

Theorem G3: field-state preservation

Suppose the initial pair is valid and each compiled operator contains a legal destination and its certified field. Selection returns such an operator. The resulting G/B relation therefore holds; modular phase remains in range; allowed metadata is reconstructed; repacking supplies correct parity and a complete mirror. Induction preserves these invariants at every accepted tick.

The manifest supplies initial node, phase and orientation and a maximum total tick count of 1..65,536. Each advance call accepts 1..4,096 ticks within the remaining budget. Budget overflow is rejected before mutation. Empty replay is permitted; a new advance of zero ticks is rejected.

The current program is a finite static geometric operator table. Changing boundary or hinge data under a new manifest changes the field and compiled operators. Runtime geometry mutation, epoch extension beyond the total budget and arbitrary program synthesis are not implemented by this profile.

Exact GPU realization

IMPLEMENTED ADAPTER | SDF.R11; ORIGINAL §14

Field construction on the device

```
D_0(v)=0 on Boundary, INF elsewhere
D_(k+1)(v)=min(D_k(v), min_u(w(v,u)+D_k(u)))
Run exactly N-1 synchronous rounds
```

Each round reads the previous buffer and writes a separate next buffer. Guarded widened integer sums preserve the chosen infinity sentinel. A shortest path has a simple representative of at most N-1 edges, so these rounds suffice. The independent certificate checks the resulting signed field before execution.

Persistent state and an integer operator texture

GPU compilation creates a 3 by N r32uint texture from the field and manifest rules. An ordered execution pass reads that immutable texture. One device invocation carries the individual's sequential state through a batch, with each tick selecting the next lookup from its predecessor. Field construction uses parallel work; the dependent state chain retains its required order.

The texture, field buffers and canonical state stay allocated across calls. Host interaction is not required between the ticks inside a batch. Host orchestration, admission, certification, archive handling and batch result transfer remain part of the implementation.

```
decode_alpha(encode_alpha(W)) = W
decode_alpha(Exec_alpha(encode_alpha(S), e)) = F_nu(S,e)
```

These are the adapter round-trip and refinement obligations. Integer bit patterns, word order and version must be preserved. Filtered sampling, color conversion and display interpolation are distinct output views, not canonical execution. Explicit GPU requests raise an error if unavailable; they do not silently evaluate the field on the CPU.

Verified: exact CPU/GPU fields and packed traces on the tested Vulkan adapter. Unmeasured: texture-cache residency, cache hit rates, device saturation, energy advantage and a general bypass of von Neumann bottlenecks.

A finite cascade of LUT stages must declare intermediate profiles, budgets and progress. Resource schedules that alter semantics are admitted replay inputs; changes only to cache residency must preserve derivation results. GPU texture semantics are described in WGSL [R1]; device profiling requires separate measurements [R2-R3].

Archives, footprints and admission

IDENTITY AND TIME | SDF.R12; ORIGINAL §§12-13

```
LUS = (nu, baseline_id, T, sequence, kind, words, status)
ID = (nu, baseline_id, producer, epoch, sequence, path)
```

LUS retains the source name Linear Unit Strain and its later telemetry role. A profile declares whether a record denotes calibrated strain, simulated boundaries, agent state or another footprint. DIGID/UUID names a structured derivation identity. Naming, representation integrity, origin admission and reproducibility are distinct results.

Arbitrary payloads are fragmented into a finite ordered sequence with total length and fragment metadata. Generated payloads may use replay descriptors when all dependencies remain available. Attribution strings are namespace metadata; they are not automatically cryptographic seeds, initialization vectors or authentication. SHA-256 in this document identifies files, not a spatial origin.

Implemented field archive

The archive retains format, complete manifest, ordered tick pairs and expected final state. Replay begins at the declared initial state and checks every retained pair. Unknown versions, extra/missing fields, malformed hexadecimal words and inconsistent replay are rejected. CPU and GPU adapters may resume each other’s archives. Adapter names and timings are diagnostics outside canonical state.

Regenerative R and forward-only W

Profile	Public interaction contract
R: regenerative	Admit named historical derivations whose original context is retained. This is the current implementation family.
W: WELip	Admit IGNITE, ADVANCE, RESIZE, INVALIDATE and EMIT; expose no historical-read operation at this interface. Formal only.

PRISM reverse lock names a specified rejection predicate for backward-history requests. “Poisoned pill” means localized active-entry invalidation with a recorded cause and retention policy. A read attempt is an event only if the interface detects it. Observation does not automatically change parity.

An internally consistent rewritten archive can still replay. Authentication needs an external trust mechanism. Atomic save and exclusive ownership protect the implemented update protocol; replay consistency is not a signature, nor a claim of immunity to storage failure.

FIFO, regeneration and finite resources

FORMAL PROPERTIES | ORIGINAL §§10, 17

```
push_C(Q,x): append x if |Q|<C; otherwise drop oldest, append x
Regen(d) = Eval_nu(omega, rules, original_T, path, inputs)
d = (nu, baseline_id, original_T, path, required_inputs)
```

Eviction removes an active derived copy. Baselines, productions, derivation descriptors and external inputs have separate retention policies. A complete pair is one eviction unit. Slot reuse must preserve derivation identity rather than confusing a new occupant with an old queue position.

Theorem R1: bounded active queue

For C at least one and initial $|Q|$ at most C , insertion below capacity increases length by one; insertion at capacity removes one item before appending one. Hence $|Q|$ never exceeds C . Reducing capacity retains the newest $\min(|Q|, C_{\text{next}})$ items and evicts older entries in order.

Theorem R2: exact regeneration

If the derivation function is single-valued and every original argument remains available, regeneration evaluates the same expression with the same arguments and yields the original word. Eviction of the active copy cannot alter that result. Missing non-derived input instead produces MISS or exclusion from the exact-regeneration set.

```
M_occupied = |Q|*w/8 <= C*w/8 bytes
M_total = M_active + M_baseline + M_rules + M_inputs
+ M_index + M_journal + M_runtime
```

A preallocated capacity- C payload arena has $M_{\text{active}}=C*w/8$ bytes, or $8*C$ bytes for 64-bit pairs. A Python cache also has container and object overhead. The bounded payload is not a bound on the full journal, descriptors or retained observations. Arbitrary external information cannot be reconstructed after its only copy is discarded.

Capacity independence and progress

When capacity changes only residency, an identical requested derivation must produce the same result at any supported capacity. Resource-driven changes to depth, resolution or selected semantics must instead be recorded as versioned input/control events. Each admitted regeneration uses a finite budget even when the architecture permits an unbounded sequence of finite requests.

Implemented: binary-world and F15 geometric-sample FIFOs, retained-context regeneration and capacity-independent decisions. Field-agent evidence includes actual eviction/reconstruction and changing capacities. Scalar fields, geometry, journals and GPU arenas remain separately accounted outside the active sample bound.

Finite Klein quotient

VERIFIED FINITE PROFILE | K1-K3

Choose strict integers W, H at least 3 with $W \cdot H$ at most 256. Temporary chart labels represent a quotient; they do not introduce a mandatory external physical origin.

```
(u, v+H) ~ (u, v); (u+W, v) ~ (u, -v)
q=floor(u/W); u0=u-q*W; v0=(( -1)^q*v) mod H
eta0=eta XOR (q mod 2); node=u0*H+v0
```

Negative labels use floor division, not truncation. Phase transported into the canonical frame is multiplied by $(-1)^q$. Node names are $k:u0:v0$ in index order. Project each unit horizontal/vertical edge, retain each sorted distinct endpoint pair once and assign unit weight. The relative seam bit τ is one on horizontal wraps between columns $W-1$ and 0 , zero elsewhere.

```
face(u, v) = [canon(u, v), canon(u+1, v),
              canon(u+1, v+1), canon(u, v+1)]
V=WH, E=2WH, F=WH; Euler characteristic=0
```

The surface claim requires cells

The audit shall check four distinct corners per face; correspondence between face boundaries and declared edges; graph connectivity; exactly two incident faces per edge; and a single cyclic vertex link at every vertex. These conditions establish a closed connected 2-manifold. Counts alone do not establish a surface.

```
x_g = -d_f*d_g*x_f
```

Here x_f is a face orientation choice and d_f is its traversal sign along a canonically oriented shared edge. Propagate the constraint through the dual graph. A contradiction establishes nonorientability. A connected closed nonorientable surface of Euler characteristic zero is a Klein bottle [R4].

At c4b41ce, KleinDomain constructs and audits this cell complex. Evidence covers all 702 admissible width/height pairs, including edge incidence, cyclic vertex links and nonorientability. Independent tests reject malformed and pinched complexes. The surface claim rests on these cells and audits, not arithmetic mirror pairs alone.

Orientation cover and seam action

VERIFIED FINITE PROFILE | K4, K7-K9

```
Cover vertices: (i,eta), encoded as 2*i+eta
(i,eta) -- (j,eta XOR tau(i,j))
(U,V) = (u+eta*W, (-1)^eta*v mod H)
```

Lift every base edge twice and each face boundary from both starting orientations. The XOR of seam bits around a face shall be zero. The cover must be connected, closed and orientable, with counts $2WH, 4WH, 2WH$ and cyclic vertex links. The displayed map identifies it with a periodic $2W$ by H toroidal grid; audit the full mapped edge and face sets, not only their counts.

A horizontal loop at $v=0$ returns to its base node after W steps with orientation reversed; its lift closes after $2W$ steps. A vertical H -step loop preserves orientation. These are explicit holonomy witnesses. The two sheets are local-frame representations of the same individual.

Packed transport

```
Operator metadata: STEP | (tau<<6)
t = (r + (-1)^eta*delta) mod 256
r_next = (-1)^tau*t mod 256
eta_next = eta XOR tau
```

Operator bit 6 stores relative seam action; operator bit 4 stays zero. Live metadata contains only STEP and orientation in bit 4. Bit 6 never leaks into live state. Set G to the destination and B to its certified scalar field, then recompute parity and the full mirror. Toggling orientation without reflecting phase is not this transport convention.

Theorem K1: mirror coherence

Let $S_{\tau}(r,\eta)=((-1)^{\tau}r,\eta \text{ XOR } \tau)$. A tick is S_{τ} composed with U_{δ} ; $M=S_1$. The phase theorem establishes U_{δ} commutes with M , and both seam maps commute. Mirrored states share G and scalar B and therefore select the same operator. Their complete tick results remain mirrors.

The proof establishes this numerical binding, conditional on correct geometry/operators. The implemented GPU compiler derives τ from the authoritative seam set and applies K8 inside its ordered tick loop; hardware traces agree with CPU execution. A generic graph with seams is not automatically a Klein surface. Orientation-cover background: Hatcher [R5].

Intrinsic ball and v2 profile

VERIFIED FINITE PROFILE | K5-K6

The implemented manifest format is relational-sdf-v2. It requires every v1 key plus seams and topology. Its archive container remains relational-sdf-machine-v1; the manifest selects execution semantics.

```
1 <= Radius <= max_v d(center,v)
sigma(v) = sign(d(center,v)-Radius)
phi(v) = sigma(v) * min_b d(v,b), b in Boundary
```

On the unit-edge quotient, every shortest path from the centre encounters each integer distance level. The selected level is nonempty and unit edges cannot skip from a negative side directly to a positive side. Recompute exact boundary distances; do not assume radial distance minus radius is an exact boundary distance on an arbitrary branched graph.

Scalar pullback theorem

The lifted field has the same value on both sheets. Every base path lifts and every cover path projects with equal length. A shortest base path to the boundary supplies a cover path to some lifted boundary point; conversely, any cover path supplies a base candidate. The two minimum distances are equal. The side function is pulled back unchanged.

v2 addition	Required interpretation
seams	Sorted unique [u,v] pairs, u<v, naming reversing edges already present in edges. Others and self transitions have tau=0.
topology	Either null for a generic relational graph or exactly {format: klein-grid-v1, width: W, height: H}.
Descriptor check	Regenerate nodes, unit edges and seams; require exact agreement with the manifest. Only audited cells establish the surface.
Compatibility	v1 retains its exact prior schema and bytes. Reject seam/topology additions to v1. Archive semantics follow the contained manifest version.

The default generator uses W=H=8, centre 0, radius 2, initial node 0, phase 250 and orientation 0. All field-class routes take local u+; increments are [11,53,137]. The measured 64-tick CPU/GPU trace crosses eight reversing seams and ends at k:0:0 with pair 11FE00D681FE002A.

Klein evidence includes negative/large labels, all 702 supported dimensions, face/link audits, connected orientable covers, full torus edge/face maps, holonomy, exact scalar pullback, CPU/GPU seam traces, replay and unchanged v1 semantics. Its 311-test baseline is preserved alongside the current 368-test result on page 27.

The infallible contract

NEW FORMALIZATION | SOLIPSISM PP.3-11

Conditional infallibility. Given a versioned profile with single-valued, total bounded transitions proved to preserve its declared invariants, a valid initial state, completely specified inputs and ordering, and faithful execution, every bounded request has one defined result; every accepted transition preserves those invariants; replay reproduces its canonical output.

```
Sigma = (States, Inputs, F, Invariant, CanonicalEncode)
I(S0)
I(S) and F(S,e)=ACCEPT(S_next,o) => I(S_next)
```

“Closed” means all output-affecting dependencies are in this profile/state/input boundary. It does not mean the executing hardware has no environment. “Infallible” names the exact contract above, not an unexplained extra operation or an assertion of physical immunity.

Theorem I1: deterministic replay

If F is single-valued, equal profiles, initial states and ordered input sequences yield equal semantic states and canonical outputs at every finite step. **Proof.** Equality holds initially. Equal states and next inputs give equal results under the same function. Induction extends equality through the finite trace.

Theorem I2: bounded total admission

If parsing is bounded, every predicate terminates, every loop consumes a finite budget and result classes exhaust the declared domain with a fixed priority, every bounded request terminates with one result. **Proof.** Each stage terminates and their finite composition terminates. Single-valued classification resolves overlaps.

Theorem I3: invariant preservation

If $I(S_0)$ holds and every accepted step preserves I , all admitted states satisfy I . **Proof.** Induction over accepted transitions. The SDF invariant includes certified field values, legal destinations, phase/metadata bounds, full mirrors and a remaining execution budget. New geometric rules require a new preservation argument.

These are mathematical implications with substantive premises. A proof of a transition theorem, verification of a program and validation against an external phenomenon are different obligations. Finite-state checking is possible; self-reference alone does not invoke an incompleteness theorem or prevent such verification.

Proof boundary and implementation refinement

NEW FORMALIZATION | SOLIPSISM PP.8-16

$$\text{decode}(\text{Exec}(\text{encode}(S), e)) = F(S, e)$$

An implementation refines a profile when this equation holds on the admitted domain, including emitted words and result classifications. A trace comparison is evidence for tested instances. It is not by itself a proof covering every possible program state, driver or device.

Layer	What must be established
Mathematical profile	Single-valued rules, total bounded admission and invariant-preserving transitions under declared premises.
Implementation	Encoding, arithmetic, memory order, admission and persistence execute those rules correctly.
Fault model	Specified corruptions are detected or recovered under an explicit physical/storage/computation fault model.
Physical interpretation	Measured units, observations and transfer functions connect the model to the represented phenomenon.

Discrete does not mean unspecified

Integer execution avoids floating-point rounding in the canonical field arithmetic. Width, signedness, overflow, shifting and conversion still require exact rules. Floating-point rounding is not inherently randomness; evaluation order and arithmetic semantics determine reproducibility [R6]. A finite discrete representation can execute an incorrect rule perfectly.

Geometry changes are accepted state changes

The addendum's internally generated growth can be formalized as a versioned graph-rewrite event. Such an event shall supply a finite rule, affected domain, retained derivation context and deterministic node correspondence. Its result must pass connectivity, boundary, range, topology and operator checks before admission. This is a contract for future growth; no dynamic SDF rewrite is implemented yet.

The source's "zero-order decay" or Lambda label can name a future deterministic invalidation function $\text{Lambda_nu}(S, e)$. Its selected entries, predicate, retention effects and result must be bound. It does not acquire a valid numerical meaning by calling unwanted input "noise." Biological reaction-diffusion analogies require actual equations before they become physical models.

Attribution remains attribution. Personal judgments in the exported discussion are not computing predicates. The meaningful architectural claim is a declared domain with exact admitted transitions, for which proof obligations and implementation measurements can be stated and discharged.

Typed zero, XOR and integrity

NEW FORMALIZATION | SOLIPSISM PP.2-3, 7, 11-16

Boolean involution

```
NOT(b)=1-b, b in {0,1}
NOT(NOT(b))=b; b XOR 1 XOR 1=b; 0*0=0
```

Two negations restore the original bit. NOT(0)=1 is one negation. If two zero operands should yield one, that is an explicitly named equality/XNOR predicate, not ordinary multiplication. Two orientation reversals likewise restore orientation; they do not establish a proposition's truth.

Zero is typed

Boolean false, arithmetic zero, phase zero, boundary distance zero, an empty container and a rejected request are different values in different types. SDF zero is a valid boundary. Rejection uses a tagged status. Clearing every bit is not automatically a valid neutral state.

```
Pair(pack(0,0,0,DATA)) = 9000000000000000
0000000000000000 fails the full RP32 mirror relation
```

Theorem I4: exact parity detection boundary

```
parity(w XOR e) = parity(w) XOR parity(e)
```

XOR associativity gives the equation. Even parity detects every odd-weight corruption mask, including every single-bit change, and misses every even-weight mask at that check. It neither locates nor corrects the changed bits. Full mirror comparison checks another relation, but a consistently replaced valid pair can still pass both checks.

```
pack(0,0,3,DATA) XOR pack(0,0,4,DATA) = 80070000
```

This result has valid parity and B=7. It is not a certificate of the distance between adjacent samples at boundary distances 3 and 4. XORing a field with itself yields zero everywhere without making every original sample a boundary. The independent geometric certificate remains necessary.

A chosen one-bit predicate can summarize an inside/boundary/outside condition, but one bit cannot losslessly encode all three classes. Binary hardware, finite packed words, geometric exactness and origin authentication therefore retain distinct contracts.

Sensory adapters and physical meaning

FORMAL INTERFACES | ORIGINAL §15; ADDENDA

Source term	Required interface binding
IPD	Relative phase between paired observations: unit, wrap, ambiguity and calibration.
IDT	Arrival/sample-time difference: clocks, units, ordering and timing calibration.
FMCW / LRAD	Sampled chirp/return model or acoustic transduction interface; deployment-specific geometry and transfer functions.
Micro-saccades	Declared small sampling or phase perturbations linked to jitter metadata.
Cicada cascades	Periodic triggers for sampling, regeneration or eviction, with a collision priority.

Delta_p = wrap_m(p_right-p_left)

Delta_t = t_right-t_left

c_i(T) = 1 if T mod period_i = 0, otherwise 0

For periods 13 and 17, both triggers coincide at multiples of 221. A reference one-bit perturbation is (c_1+c_2) modulo two. The manifest must specify its effect on phase or sampling and recompute parity after an intentional state change. Jitter is not an integrity fault by definition.

Observed samples, calibration versions and semantic scheduling decisions belong to the replay context. Converting them to range, direction, strain or density requires a physical signal model. Inverse-square physics does not follow from logarithmic addressing; wave mixing is not automatically Hadamard lane multiplication or XOR.

Applications permitted by the contract

The architecture can host deterministic simulation, regenerable world models, compact geometric controllers and device-independent continuation once their profiles and adapters are specified. Robotics, optical/FPGA execution, biological growth and physical autonomous construction require additional realizations. The current repository demonstrates software/GPU substrates and a simulated/live-observation individual.

A complete application must identify what the field means, where its inputs come from, what action an output authorizes, and how the model is validated. The formal architecture supplies a common representation and execution discipline; it does not supply an absent sensor or actuator model.

Implementation map and integration

MEASURED PROFILES | FI1-FI8

The field-agent policy joins recipe-derived Klein geometry to the existing individual. The module map identifies implemented scope; remaining architectural concepts retain their separate obligations.

Layer	Existing implementation	Scope
Packed core	rp32.py	Encoding, phase, parity, full mirror and pair.
Derived world	world.py; runtime.py	Binary derivation, bounded active FIFO, replay and demo archives.
Individual	motion.py; navigation.py; tomigidt.py	Costs, incremental deterministic planning, local observations and action state.
Persistence / live	session.py; live.py	Ownership, atomic saves, admitted external frames, deduplication and durable acknowledgment.
Packed GPU	gpu.py; packed.wgsl	World derivation and selected motion forecasts; CPU retains planning and admission.
Intrinsic field	field.py; field_cli.py	Strict geometry/manifest, exact field, operators, one persistent sequence and replay.
Field GPU	sdf_gpu.py; field.wgsl	Device field construction, operator texture and ordered state transitions.
Klein quotient	klein.py	Audited cells, connected orientation cover, intrinsic ball and v2 seam transport.
Field integration	field_agent.py; field_world.py; field_agent_gpu.py	Typed agent recipe/state, sample FIFO and persistent device actions; field_agent.wgsl.

Profile meanings remain explicit

Earlier motion/world profiles use B for terrain or energy; relational-sdf-v1/v2 use B for exact signed distance. FI1-FI8 preserve these schemas and implement a field-agent policy with B=phi and separate integer energy. Shared encoding never permits an implicit role change.

Measured integration

One existing Tomigidt plans over recipe-derived geometry, observes local hazards, transports its state through seams and regenerates samples. CPU/GPU canonical histories agree. Psi/f8, general growth and physical adapters remain distinct obligations.

Paths are relative to solvefinite/. Integration source hashes and command provenance: docs/evidence/field-agent-v1/. Retained Klein baseline: docs/evidence/klein-field-v2/. Formal-first identity and chronology are on page 33.

Fresh verification and field trace

OBSERVED RESULTS | 25 SEPTEMBER 2026

Measurement	Observed result
Full integrated suite	368 tests passed in 36.348 s; zero skipped.
Actual device coverage	41 GPU methods: the prior 28 plus 13 field-agent methods.
Klein topology coverage	All 702 admissible dimension pairs audited, including base/cover surfaces and full torus edge/face maps.
Conformance examples	Klein: 22 checks; field agent: all 23 checks passed, including field ablation, FIFO reconstruction and replay.
Hardware / software	NVIDIA GeForce RTX 5070 Ti Laptop GPU; Vulkan; driver 591.59; wgpu 0.32.0; Python 3.12.14.
Cross-adapter continuation	CPU/GPU archives agree; fresh processes replay both ways before the seam and during DEFER. Live retries preserve history.

```
SDF v1: [-6,-4,-3,0,2,3,5]
Moved boundary: [-8,-6,-5,-2,0,1,3]
V1 tick 64: n4, pair 1102046C01020494
Klein v2 tick 64: k:0:0, pair 11FE00D681FE002A
```

Changing geometry changes field execution and agent route selection. The FI8 mission completes after four cycles at pair 06011145160111BB, energy 90. Its reversing seam yields phase 240 and eta 1. The bounded-search mission takes 41 cycles; six energy/recipe/valid-parity forecast tamper probes are rejected.

Reproduce the measured checks

```
python -m unittest discover -s tests -v
python -m examples.field_agent_conformance --output agent.json
python -m examples.klein_conformance --output klein.json
python -m examples.field_conformance --output sdf.json
```

The historical SDF capture (262 tests/21 device methods) and Klein capture (311/28) remain retained evidence, including the seven-node boundary ablation above. Counts describe tested finite profiles, not an architectural completion percentage.

Current evidence: docs/evidence/field-agent-v1/. Earlier captures: docs/evidence/klein-field-v2/ and docs/evidence/formal-edition-2026-09-25/. Logs, source hashes, traces, adapter details and complete archives identify each observation.

Progress against the architecture: I

CAPABILITY STATUS | FINITE REALIZATIONS

Obligation	Measured status and remaining work
C1-C3: relational packed execution	Verified subset. RP32 and state-selected integer texture execution work. General active log-radius scale transitions and alternate historical carriers remain unbound.
Exact intrinsic SDF	Verified subset. Weighted graph distance, declared separating boundary, units, certificate and field-governed traces work. Continuous primitives and physical calibration are separate.
C4: Klein geometry	Verified finite profile. Quotient cells, nonorientability, connected orientable cover, torus maps, intrinsic ball and CPU/GPU seam transport are audited.
Psi traversal	Supplied routes verified. An eigenvector-derived traversal needs operator, eigenvalue selection, normalization, degeneracy and frame transport.
Hadamard / Delta-Delta	Partly formal. Typed arithmetic and field bounds are defined; second-difference bounds are checked. A geometric Hadamard routing profile is not implemented.
f8 middle-out index	Formal / unbound. Total key, canonical identity and median recursion are stated. Full canonical ordering, tie policy and versioned rebuild are not implemented.

What the tests establish

The implemented graph profile has exact arithmetic and a field certificate, and its tested CPU/GPU realizations agree. These are substantial completed components. They do not establish every named architectural layer by association with the same word carrier.

What will count as progress next

FI1-FI8 now have end-to-end field-guided planning, observation-driven replanning and genuine sample eviction/regeneration evidence. Finite Klein cells, covers, holonomy and CPU/GPU transport remain verified. Psi/f8/Hadamard still require explicit numerical bindings and independent conformance checks.

The requirements matrix continues on page 29. No percentage is assigned because architectural obligations differ in size and some remain unbound; counting passing tests would produce a misleading completion estimate.

Progress against the architecture: II

CAPABILITY STATUS | INTEGRATION AND APPLICATIONS

Obligation	Measured status and remaining work
C5: generative world	Verified subset. Binary derivation and recipe-only Klein cold rebuild work. General primitives and geometry-changing grammar execution remain separate.
C6: finite active memory	Verified subset. Pair-atomic sample FIFO, genuine reconstruction and capacity-independent agent histories work. Journals, geometry, fields and device arenas consume additional memory.
C7: complete mirror	Verified subset. Phase involution/commutation, parity and full pair relations work, including K8 phase reflection and orientation transport on CPU/GPU.
C8: individual and footprint	Verified subset. Planning, observations, epoch/sequence, retry handling, ownership and durable continuation work. Full LUS/DIGID envelopes and global admission/authentication remain profile obligations.
WElip / Lambda	Formal / unbound. Current profiles permit regeneration. A forward-only interface and complete invalidation/clock-wrap binding remain to be implemented.
GPU / performance	Verified subset. Exact tested adapters and device execution exist. Cache saturation, general speed/energy superiority and other hardware adapters are unmeasured.
Waves / physical growth	Unbound applications. Signal models, physical transfer functions and calibrated action semantics are not supplied by the present code.

Continuation after the integrated field agent

FI1-FI8 join retained geometric recipes to the existing individual’s observation/planning loop, with legacy fixtures preserved. Remaining work binds traversal/index operators, geometry-changing growth and wider continuation. Each stage requires evidence against its own obligations.

The measured integration is finite and profile-scoped. Whole-system indefinite continuation, autonomous physical self-replication, consciousness and universal immunity are not demonstrated outcomes of the present profiles.

Conformance and performance scope

ACCEPTANCE CRITERIA | ORIGINAL T1-T10

ID	Criterion and present scope
T1-T4	Exact encode/decode, modular phase, mirror involution/commutation and odd-bit parity detection: exercised for RP32.
T5	Canonical topology and transported field meaning: finite Klein K1-K9 are implemented and independently audited.
T6-T8	FIFO order/pair atomicity, retained-context byte replay and explicit profile/version handling: exercised for existing profiles.
T9	Equal semantic histories under tested capacities/batches/adapters: exercised in finite implementations; no general distributed deployment proof.
T10	Declared application outputs: graph-field and simulated/live-observation contracts exercised; physical calibration remains unbound.

Historical substrate benchmarks

Workload	Logical transitions	Measured batch wall time
Depth 16, 65,536 paths, 20 repeats	20,971,520	0.0018765 s; about 11.18 billion/s
Depth 18, 262,144 paths, 500 repeats	2,359,296,000	0.0275231 s; about 85.72 billion/s

These repository records date from 25 September 2026 and concern the packed binary substrate. They repeatedly use a 32-byte LUT. Timings include submission/readback but exclude initialization, compilation and upload. Readback sizes were 1,048,576 and 4,194,304 bytes; explicit GPU payload accounting was 2,098,264 and 8,389,720 bytes. CPU results agreed for the benchmark cases.

They are not fresh end-to-end autonomy timings, optimized CPU comparisons, cache-hit measurements or proof of device saturation. Logical transition rates for these particular loops cannot be generalized into an architectural bandwidth claim.

Required comparison record

A meaningful performance comparison records hardware, profile, rule set, LUT and working-set sizes, batch distribution, precision, initialization, transfers, retained inputs, total memory, latency, throughput, regeneration cost and output agreement. Compare the same workload and correctness target. Cache and occupancy claims need corresponding counters [R2-R3].

Historical records: docs/benchmarks/rtx5070ti-depth16.json and rtx5070ti-depth18.json. Fresh correctness results are on page 27. No new performance benchmark was run for this PDF detour.

Executable default and archive schema

SELF-CONTAINED REFERENCE | SDF.R1-R12

This complete default manifest reproduces the implemented seven-node field. All three route columns must be legal even when the current field selects only one.

```
{
  "format": "relational-sdf-v1", "identity": "TOMIGIDt",
  "nodes": ["n0", "n1", "n2", "n3", "n4", "n5", "n6"],
  "edges": [[0,1,2],[1,2,1],[2,3,3],[3,4,2],[4,5,1],[5,6,2]],
  "signs": [-1,-1,-1,0,1,1,1],
  "routes": [[1,1,0],[2,2,0],[3,3,1],[4,4,2],
             [5,5,3],[6,6,4],[6,6,5]],
  "turns": [[11,53,137],[11,53,137],[11,53,137],
            [11,53,137],[11,53,137],[11,53,137]],
  "unit_num": 1, "unit_den": 1,
  "initial_node": 0, "initial_phase": 250,
  "initial_orientation": 0, "max_ticks": 65536
}
```

Reference evaluation

```
D = exact_shortest_distances_from_all_boundary_nodes(manifest)
phi = tuple(sign[i] * D[i] for i in range(N))
certify_field(manifest, phi)
W = pack(initial_phase, initial_node, phi[initial_node],
         STEP | (initial_orientation << 4))
for tick in admitted_finite_budget:
    r, i, b, a = unpack(W)
    c = 0 if b < 0 else (1 if b == 0 else 2)
    delta, dest = turns[i][c], routes[i][c]
    r = (r + (-1 if a & 16 else 1) * delta) % 256
    W = pack(r, dest, phi[dest], a)
    emit(Pair(W))
```

Canonical archive

Top-level keys are `format`, `manifest`, `trace`, `expected`. Format is `relational-sdf-machine-v1`. Trace entries are exactly 16 uppercase hexadecimal characters, one per accepted tick. Expected contains `identity`, `tick`, `node`, `agent_pair`. The manifest reconstructs the initial pair; trace length is the tick count. Every entry and expected final value must agree with replay.

The JSON schema is strict: unknown/missing keys are invalid. Canonical word equality does not require incidental JSON whitespace to match. The initial pair is not an emitted tick. State ownership and atomic file replacement belong to the CLI persistence adapter, not to a distance equation.

Required bindings and stable obligations

PROFILE COMPLETENESS | ORIGINAL APPENDIX D

Binding family	Required decisions / original requirement IDs
Dependencies	Baseline, version, inputs, deterministic interpretation and retention; original R1.
Phase / scale	OTAN2 increment, miss/tie/endpoint behavior, log unit/base, resolution and budget; R2.
Topology / geometry	Canonical representatives, seam/mirror action, metric, boundary, signs, primitives and units; R3-R4.
Words / grammar	Layout/version, continuation; rule priority, branches, original time and finite derivation; R5-R6.
Index / routing	Canonical total keys, ties/rebuild; typed Hadamard, gradient, widths, rounding and admissibility; R7-R8.
Regeneration	Required non-derived inputs, capacity policy, pair atomicity, missing-input and budget results; R9-R10.
Identity / time	Epoch/input prefix, event ordering, wrap, identity allocation and separate integrity/origin outcomes; R11-R12.
Forward interface	Detected-event predicate, localized invalidation and unaffected retained partitions; R13.
Adapters / scaling	Exact word preservation, finite cascade stages, ownership, input delivery and resource controls; R14-R15.
Physical adapters	Samples, units, calibration, transfer functions and retained scheduling effects; R16.

Original R1-R16 are preserved above. Implemented extensions use **SDF.R1-R12** on pages 13-17 and 31, **K1-K9** on pages 19-21 and **F11-F18** on pages 35-38. Their scopes remain separate; satisfying one profile does not silently complete another.

Original property cross-reference: P1 replay is I1 on p.22; P2 FIFO and P3 regeneration are R1/R2 on p.18; P4 mirror preservation is on p.8; P5 representation independence is the adapter/refinement contract on pp.16,23.

Historical vocabulary: mosTADPOLE(thegreenone) names the source input/output lineage; TPVM is Topological Fixed-Point Virtual Machine. Det-0 names reproducibility with complete dependencies. Generative Topological Fabric, Packed Topological FIFO Matrix and Uniform Packed Operator Paradigm are successive functional views. REW expands to Reverse Electronic Warfare in the WELip source. f8's "BVH replacement / S2 superseder" is an intended indexing/comparison role, not a measured replacement result.

A complete manifest may reference immutable versioned rule tables rather than embedding every algorithm in a packet. No "infallible" designation waives a missing binding. A profile is executable only once all choices that affect its outputs are single-valued and finitely evaluable.

Source concordance and provenance

RETAINED CORPUS | CONSOLIDATED READING EDITION

Source	Integrated destination in this edition
O: TK-LPLUT-1.0, 24 pages	Original §§1-3: pp.4-6; §§4-5: pp.11-14,19-21; §§6-8: pp.7-10; §§9-11: pp.9,11,18; §§12-16: pp.5,16-17,25; §§17-18: pp.8,14,18,22-24,27-30. Appendices A-B: pp.8,31; C-D: pp.32-34.
S: Solus, 7 pages	pp.3-4: operator geometry and hinge/traversal roles; pp.5-6: relational fields and self-reference. Integrated pp.6,11-16,24-25. Search/conversation material is not a new protocol.
I: solipsism, 16 pages	pp.3-5: one closed individual, pp.9-13: infallibility/SDF claims, pp.15-16: propagation/decay intent. Integrated pp.9,12,22-25. pp.2-3 arithmetic is corrected in typed form on p.24; personal analogies on pp.14-15 supply no predicates.
SDF companion / Klein companion	All normative SDF.R1-R12 and K1-K9 are consolidated in this PDF. Earlier Markdown remains the recorded development sequence.

SHA-256 of preserved source PDFs

O 8ea9cfb077630993e1d472ba72715a25d
2402bf243518663f7d08b03f8b83647
S a2ace794138bcad68463d1ce2ba4a6e99
54540bb4f2f29e4090d68d6e0440489
I 1091e0435aa1648b4ede127837e66ddbfb
2c5dbd0a67f49cbb7a7ae134ee547e7

Join each source's two lines without spaces. Retained paths: O at repository root, S at sources/solus-ion-ad-infinitum.pdf, I at sources/solipsism.pdf. Both addenda originated in the author-named "Philosophers stone Jitske Klootwijk" source directory.

Implementation evidence identity

Klein: c4b41ce12a33a747bd54c8cc7f9748a06f7b59de
Capture: 2026-09-25, 11:30:28 UTC
Prior: 8f4b87bed131f5c084ef59aec558f3f9eb6ddedc
Prior capture: 07:52:57 to 07:54:16 UTC

Chronology: SDF specification 477e576 preceded implementation 50c9389; Klein specification 8f4b87b preceded c4b41ce. FI1-FI8 revision 1 was committed as 5ccc022 before integration runtime coding. Revision 2 binds measured integration source hashes in docs/evidence/field-agent-v1/verification.json. All earlier captures and original source bytes remain preserved.

Notation and technical references

REFERENCE INDEX

Symbol / name	Meaning
nu; omega; T	Version; retained baseline; logical tick.
r; eta; delta; tau	Phase; local orientation bit; phase increment; relative seam bit.
phi; D; sigma	Signed field; unsigned boundary distance; declared side.
Psi; theta; f8	Bound traversal/eigenvector role; local sweep direction; relational index.
M; Q; C	Full mirror; active FIFO; active capacity.
RP32; LUS; DIGID	Packed carrier; typed emitted footprint; structured derivation identity.
R / W; Lambda	Regenerative/forward-only interaction profiles; proposed bound invalidation function.

Primary technical references

[R1] W3C. [WebGPU Shading Language](#), [textureLoad](#). Integer texture access semantics. This reference supplies API background, not a proof of this runtime.

[R2] NVIDIA. [CUDA C++ Best Practices Guide](#), version 12.8. Memory hierarchy and measurement background; the tested adapter here uses Vulkan through wgpu.

[R3] NVIDIA. [Nsight Compute Profiling Guide](#). Profiling methodology and counters. No Nsight counter evidence is claimed in this edition.

[R4] Jacob Lurie, MIT. [Classification of Surfaces](#), [Lecture 33](#). Surface classification used conditionally after manifold/orientation checks.

[R5] Allen Hatcher. [Algebraic Topology](#), [section 3.3](#). Orientation covers. The concrete finite quotient and required audits are specified in this edition.

[R6] David Goldberg. [What Every Computer Scientist Should Know About Floating-Point Arithmetic](#) (1991). Rounding and evaluation semantics; cited to separate numerical error from inherent randomness.

[R7] John C. Hart. [Sphere Tracing: A Geometric Method for the Antialiased Ray Tracing of Implicit Surfaces](#) (1996). Context for geometric distance bounds. The present certificate is proved directly for a finite graph.

Revision 3 status. SDF, Klein and FI1-FI8 retain their measured finite evidence. Pages 39-43 add the formal-first PX1-PX8 binding; its runtime is pending. The prior 368-test, 41-device-method and 23-check integration capture establishes no PX runtime claim. Wider architecture remains separately scoped.

One field-guided Tomigidt

VERIFIED FINITE PROFILE | FI1-FI2 | FORMAL-FIRST CONTRACT

FI1. Identity, schema and typed state

FieldAgentManifest shall select policy tomigidt-field-observe-plan-act-v1, word_profile RP32-relational-sdf-v2 and perspective local-observation-v1. One existing Tomigidt owns the goal, local observations, retained search, live state and event history. There is no separately advancing FieldMachine or second individual.

The exact manifest keys shall be those below. Unknown/missing keys, Boolean integers and invalid shapes shall be rejected. JSON arrays become immutable tuples. Target and initial_node are canonical names; initial_node defaults to k:0:0. The .start property supplies that same name. No graph, routes, field array or sensor-supplied phi belongs in this schema.

identity, target, world, initial_node, initial_phase,
initial_orientation, initial_energy, repair_cost,
max_search_expansions, max_hops, max_cycles,
policy, word profile, perspective

```
Live state = (Pair(pack(r,i,phi(i),STEP | (eta<<4))), E)
0 <= E <= 2^31-1; E is separate from the B lane
```

Phase is 0..255 and orientation is 0 or 1; defaults are 250 and 0. Initial energy defaults to 100. Repair cost is 1..127; cycles are 1..1,000,000; search quantum is 1..65,536 expansions per cycle. Identity is a nonempty name of at most 128 characters. All numerical bounds use strict integers. Completion uses EMIT instead of STEP while retaining orientation.

FI2. Retained geometric recipe

World recipe format is klein-ball-world-v1 (Python attribute version). Its exact JSON keys are format,width,height,center,radius,turns,baseline_id. Baseline ID is a nonempty string. Defaults are W=4, H=5, centre index 0, radius 2 and turns [11,53,137]. Width/height obey K1; centre is in 0..WH-1; radius obeys K5; turns are three unsigned eight-bit integers.

The retained recipe regenerates canonical k:u:v nodes, every unit quotient edge, seams, cells and exact certified SDF. It replaces an expanded graph/field in the manifest. K1-K5 and the scalar mirror rule remain binding. Geometry is immutable within one manifest; original external observations remain separate retained inputs.

This policy is a new typed application binding. Legacy AgentManifest policies, terrain/energy words, FieldMachine schemas, archives and fixture bytes retain their previous meanings. The shared session/live envelopes select this policy from their contained manifest.

Observe, plan and transport

VERIFIED FINITE PROFILE | FI3-FI4 | ONE ACTION SEQUENCE

FI3. Local observations and retained planning

Sensors supply only strict hazards in 0..127 for the current node and its outgoing quotient neighbors. DATA observation packets retain B=hazard under their explicit observation context; they are not SDF samples. No input may supply a route, action or phi. Before MOVE or REPAIR, require a fresh complete local frame. Partial frames produce WAIT; unseen distant nodes use the existing zero-hazard hypothesis.

Generate the graph from all quotient edges. Use global lexical ordering of canonical names and lexical route ties. RouteSearch gains explicit relational mode with a strict hop bound in 1..255, default 255; legacy binary mode retains its 32-hop bound. Per-cycle search quanta retain unfinished work through DEFER. Search context includes position, current pair, separate energy and effective costs; a changed context invalidates stale work before an action can be admitted.

```
c(i,j,h_j) = 1 + abs(phi(j)) + h_j
route_cost = sum c over its entered nodes
MOVE requires route_cost + repair_cost <= E
```

This cost is a declared application rule in integer codes, not physical energy calibration. The field shall affect actual route selection. Forecast the internally selected route without mutating canonical live state. Empty forecasts are allowed but shall still validate the seed. Real moves must follow adjacent distinct nodes; target handling uses REPAIR, not a stationary movement.

FI4. Exact packed action and target repair

For admitted i to j, source phi selects the negative/zero/positive increment from recipe turns. Derive tau from the authoritative edge seam set. Apply the increment in the departure frame, then K8 reflection and orientation transport. Require exact destination phi, recomputed parity and a complete mirror.

```
delta = turns[class(phi(i))]; tau = seam(i,j)
t = (r + (-1)^eta * delta) mod 256
r_next = (-1)^tau * t mod 256; eta_next = eta XOR tau
w_next = pack(r_next,j,phi(j),STEP | (eta_next<<4))
E_next = E - c(i,j,h_j)
```

At the freshly observed target, require E at least repair_cost. REPAIR preserves R, G, B and eta, changes the opcode to EMIT, subtracts repair_cost from separate energy and marks COMPLETE. A negative or zero field never means negative or zero energy. Decisions, forecasts, live pairs and energy shall be checked as one admitted history.

Planner-selected destinations are this policy's own internal decisions. FI4 does not replace or reinterpret the fixed per-class routes of standalone FieldMachine. Existing observation admission, event ordering and finite-cycle outcomes remain in force.

Regeneration and device admission

VERIFIED FINITE PROFILE | FI5-FI6 | RESIDENCY AND HISTORY

FI5. Recipe-derived samples and bounded FIFO

```
sample(i) = Pair(pack(0,i,phi(i),DATA))
Regen(recipe,i) = the same certified sample(i)
```

The geometric world shall supply pure derive and caching get. Derive changes no cache order, counters, observations or live state. On CPU it recomputes from the retained recipe; on GPU it re-materializes the sample from the certified field buffer. Get stores atomic complete pairs in a capacity-C FIFO, C a positive strict integer. Hits do not refresh insertion order; overflow and shrinking evict oldest insertions. A later miss for an evicted node must actually reconstruct the sample.

Cold restart rebuilds geometry, exact fields and required operators from the recipe. Regeneration retains baseline/version and original dependencies; it never substitutes a later context. Hazards and other admitted measurements must survive in the observation journal. They cannot be reconstructed from the geometric recipe. Missing original dependencies shall fail admission, not silently synthesize measurements.

```
active sample payload <= 8*C bytes
M_total = M_FIFO + M_recipe + M_fields + M_operators
+ M_geometry + M_observations + M_journal
+ M_search + M_runtime
```

The full scalar field tuple and GPU scalar/operator arenas are accounted outside the FIFO; the bound does not describe total memory. Python objects also have overhead. Cache capacity and incidental eviction affect residency only and stay outside canonical state. Forecasting must not admit unobserved measurements or populate the observed active sample cache.

FI6. GPU execution with one persistent owner

The device shall construct the field; an independent geometric certificate shall admit it before device operators are used. Compile every allowed neighbor/class operator from certified fields and authoritative seams: four distinct neighbors times three classes, a 12 by N Klein arena. Operators carry delta, destination, destination phi and relative seam bit; live metadata shall never inherit operator bit 6.

Forecast a caller-internal planned route of at most 255 edges in scratch device state without changing the live owner. Accepted MOVE and REPAIR shall dispatch from persistent GPU pair and energy, not upload a host-computed forecast as the action. Compare the device result with the admitted prediction before committing the journal. An uncertain device outcome or prediction mismatch fails closed; reconstruct from the last durable admitted history before further work.

CPU/GPU fields, costs, forecasts, decisions and canonical archives shall agree exactly. Explicit GPU execution shall not fall back to CPU field or transition evaluation. Adapter names, timings and cache diagnostics remain outside canonical state; host planning and admission remain explicit host stages.

Integrated replay and acceptance

VERIFIED FINITE PROFILE | FI7-FI8 | CONFORMANCE

FI7. One journal, durable admission and replay

Reuse Tomigidt’s session/live envelopes, process locks, atomic saves and durable acknowledgment. Field-policy event keys shall be exactly seq,input,decision,output,energy; energy is a strict integer. Snapshots include energy and word_profile; retained planning summaries include energy in their context. Legacy events retain their exact four-key schema.

Replay regenerates geometry and re-admits original observations in order, recomputing each (pair,energy) state rather than installing stored energy. Check every complete event, forecast and expected state, including WAIT/DEFER and search progress. Reject supplied replacement recipes for retained sessions and all replay-inconsistent records. Backend/capacity changes shall preserve canonical archives. Invalid frames shall not mutate live state, energy, search, observations or cache; legacy fixtures remain exact.

FI8. Reference mission and independent checks

Use the 4 by 5 default recipe, start k:0:0, target k:3:2, phase 250, eta 0, energy 100 and repair cost 5. Hazards are zero until hazard 70 at k:0:3 takes effect from cycle 2, and only visible hazards are reported. Initial lexical least-cost route is [4,3,17], cost 5. After entering node 4, the new local observation selects [16,17], cost 3. Edge 4 to 16 is a reversing seam.

Admitted action	Node / phase / phi / eta	Separate energy
MOVE 1	4 / 5 / -1 / 0	98
MOVE 2, seam	16 / 240 / 0 / 1	97
MOVE 3	17 / 187 / +1 / 1	95
REPAIR, EMIT	17 / 187 / +1 / 1	90

With zero hazards and only the centre changed to index 4, the first planned route becomes [15,16,17], cost 4. This ablation proves field geometry changes planning, rather than decorating an unrelated route. Both CPU and GPU now reproduce these formal-first reference vectors and complete canonical archives.

Required checks. Strict recipe/manifest and typed lanes; preserved legacy formats; independently calculated costs and packed words; field ablation, hazard replanning, seam reflection and mirrors; FIFO hit order, real eviction/reconstruction and recipe-only cold rebuild; capacity-independent histories; stale-search invalidation; fresh-process and both-backend replay; device action ownership, no fallback and fail-closed errors. Energy tampering shall fail complete-event replay.

All 23 field-agent conformance checks pass; the full suite contains 368 tests, including 41 actual-device methods. FI1-FI8 do not claim eigenvector Psi, full f8, geometry-changing growth, infinite memory, calibrated physical energy or completion of the whole architecture.

Local SDF-derived Psi binding

FORMAL PENDING | PX1-PX2 | NUMERICAL CHOICE BEFORE RUNTIME

PX1. A declared local eigen-operator

This profile makes a new explicit numerical choice for the eigenvector/traversal and f8 roles in the original source (original pages 7 and 11; consolidated page 11). It is not asserted to be the unique formula implied by that source. Its operational role is indexing recipe-derived scalar base descriptors. Hadamard routing, a global Psi operator and behavior-changing traversal remain separate bindings.

Use the immutable Klein recipe, canonical node $i=uH+v$ and certified unit-edge scalar field ϕ . Directions $u+$, $u-$, $v+$, $v-$ are the K1 quotient directions in the canonical chart. Define the unscaled symmetric difference g and its positive-semidefinite local operator exactly:

```
g_u = phi(u+) - phi(u-); g_v = phi(v+) - phi(v-)
g = (g_u, g_v); A = g g^T
g != 0: lambda = g_u^2 + g_v^2
P = g / gcd(abs(g_u), abs(g_v))
g == 0: A = 0; lambda = 0; P = (1, 0)
Psi = epsilon * P; epsilon in {-1, +1}
```

For nonzero g , λ is the unique largest eigenvalue, and $A*P=\lambda P$. Integer gcd normalization and the declared sign select a nonzero primitive axis. For $g=0$, all vectors lie in the zero eigenspace; the explicit $P=(1,0)$ tie rule chooses the axis. The fallback is a declared convention, not a recovered physical direction.

PX2. Bounds and chart transport

The certified unit-edge SDF has adjacent differences at most one. Thus both components of g lie in $[-2, 2]$. Gcd-normalized P and either Ψ sign also lie in $[-2, 2]$; λ belongs to $\{0, 1, 2, 4, 5, 8\}$. Use exact widened integer arithmetic; no floating-point eigensolver, tolerance or rounded normalization participates.

```
J = diag(1, -1)
g_prime = J*g; A_prime = J*A*J
Psi_prime = J*Psi
```

These transformations bind an orientation-reversing Klein chart. Transport the chosen axis, including the declared degenerate choice, through J . Keys shall be derived only in the canonical frame. A temporary label or opposite local frame changes displayed components, not the indexed scalar identity.

PX runtime is pending. Mathematical checks of these formulas and formal reference vectors are not evidence that a CPU/GPU index, lookup path or rebuild has been implemented.

Canonical index identity and key

FORMAL PENDING | PX3-PX4 | ONE BASE DESCRIPTOR PER NODE

PX3. Immutable binding and derivation identity

IndexBinding has exactly the JSON keys below. Reject missing/unknown keys, Boolean integers and invalid values. Format is f8-klein-sdf-v1; epoch is a strict integer in $0..2^{31}-1$, psi_sign is exactly -1 or +1, and phase_origin is in $0..255$. Defaults are epoch 0, sign +1 and origin 0.

format, epoch, psi_sign, phase_origin

Full version identity is the complete Klein recipe content together with this binding, including baseline ID, geometry, centre/radius and turns. Epoch is scoped to that lineage; it is not a globally unique identity. Index all N recipe-derived scalar base descriptors, independently of FIFO residency. Oriented live agent states are not additional entries.

```
canonical derivation address i = u*H + v
d(i) = unweighted quotient distance from recipe.center
parent(root) = root
parent(i) = min numeric j among neighbors with d(j)=d(i)-1
```

The parent rule selects one finite root path. Numeric node order here differs from F13 lexical name/route order. G remains the stable canonical node identity throughout; it is never replaced by a tree row. Alias queries reduce signed temporary Klein coordinates to their one canonical scalar identity. Strict k:u:v parsing in existing manifests remains unchanged.

PX4. Radius, phase and the total key

Define rho by integer bit length. It buckets graph radius plus one, and does not implement an active physical log-scale transition. Begin the root derivation with phase_origin and eta=0. Advance the K8 increment/reflection rule along the unique parent path, using each departure node's field class to select its recipe turn.

```
rho(i) = floor(log2(d(i)+1)) = bit_length(d(i)+1)-1
theta(i) = (-1)^eta * r mod 256
          = (phase_origin + sum departure turns) mod 256
key(i) = (Psi_u+2, Psi_v+2, rho, theta, i)
```

Each key has five unsigned 32-bit components. Compare them lexicographically by numeric component, never by string representation or a hash. The final canonical node ID guarantees uniqueness even when earlier components tie. A different sign or phase origin may reorder storage while preserving every geometric node and movement edge.

The phase derivation above is recipe/index metadata. It does not install a new phase in Tomigidt, append a movement, consume energy or supply a route to the planner.

Middle-out tree and executable lookup

FORMAL PENDING | PX5-PX6 | STORAGE ROWS AND DEVICE USE

PX5. Exact lower-median tree

Sort all canonical node keys. For each half-open sorted interval $[lo,hi)$, choose its lower median; emit that node, then its left and right subtrees recursively. Physical tree rows are this preorder. The resulting row layout is canonical for the complete recipe and binding.

```
mid = floor((lo + hi - 1)/2)
left interval [lo,mid); right interval [mid+1,hi)
tree row = [key0,key1,key2,key3,key4,leftrow,rightrow,0]
node-key record = [key0,key1,key2,key3,key4,lambda,g_u+2,g_v+2]
```

Both record types contain eight u32 values. N is 9..256; null child is 256 and the root is row 0. A valid search takes at most $\text{bit_length}(N)$, at most nine, visited rows. Exact key equality returns the stored row and canonical node identity. A well-formed absent key returns MISS without any state change. Malformed rows, links, ordering, duplicate/missing identities or trees are invalid, not a successful miss.

Tree edges are storage-search relations and shall never be interpreted as movement edges. The geometric graph, field, sensor neighborhood and canonical node identity remain authoritative. CPU world derive shall perform the actual tree walk before freshly regenerating a sample; a decorative tree or a direct node-to-row bypass does not satisfy this requirement.

PX6. Certified construction and actual GPU dispatch

The GPU shall construct keys, Psi values, numeric ranks and the complete tree from certified device fields and immutable geometry, including directional neighbors and parent/depth data. No host key compiler may substitute for explicit GPU construction. Node-key records and all rows shall exactly match independent CPU results.

Compile the neighbor/class operator texture in tree preorder. The shader shall obtain a query key from the current state's canonical G and walk the tree to the actual operator texture row. Validate the returned node identity before using its operator. This lookup must govern actual device actions as well as forecasts; G itself remains the canonical node ID.

Admission certificates shall validate field-derived g, A/eigenvalue, gcd normalization, sign, parent-distance relation, K8 derivation phase and every key. They shall also validate the complete canonical lower-median/preorder tree and its node coverage. Parity or a locally plausible child link alone is insufficient.

Existing FI6 persistent state, no-CPU-fallback, prediction comparison and fail-closed requirements remain binding. A corrupted row or wrong returned identity shall not silently select another node's operator.

Atomic rebuild and semantic preservation

FORMAL PENDING | PX7 | VERSIONED STORAGE ALTERNATIVE

PX7. Prepare, certify and swap one version

A rebuild creates a new immutable IndexBinding with epoch incremented by exactly one; psi_sign and phase_origin may optionally change within their declared bounds. Reject epoch overflow at $2^{31}-1$ before mutation. This is the versioned alternative allowed by original requirement R7, not an in-place ambiguous reordering.

Prepare and validate the complete candidate keys, tree, operator texture and device bind-group bundle before a serialized atomic swap. Each lookup/action uses one complete admitted bundle. Retain enough ownership to distinguish a rejected candidate from an already committed replacement.

Outcome	Required effect
Invalid input or pure certificate failure	Reject with old index, canonical state, active FIFO/order/counters and retained search unchanged.
Uncertain device outcome	Close the owner and retain unchanged admitted history; recover from that history before further work.
Validated candidate, atomic swap	Commit the complete new bundle while preserving semantic history and working residency.
Cleanup failure after the swap	Report a committed cleanup failure. Do not report rejection or pretend the old version remains active.

Storage refinement, not a planning event

Keys and tree order shall not determine planner iteration, costs, lexical ties or search quanta. Rebuilds preserve all agent pairs, separate energy, events, observations, DEFER frontier/cursor and active FIFO contents, order and counters. Residency does not change merely because its descriptor has a new tree row.

Expose the complete binding and version identity in diagnostics. Replay may use a different independently verified index version and must produce the same canonical archive. Do not add incidental row order or index epoch to legacy/field-agent semantic schemas. If a future Psi choice changes route behavior, bind a new semantic version or an admitted event first.

```
M_total_next = existing FI5 accounting + M_index
M_peak_rebuild includes old bundle + candidate bundle
M_index includes all retained descriptor/key/tree metadata
```

Index metadata is separate from the bounded FIFO. It shall not hide a complete arena of packed world-node pairs. Account for both old and candidate allocations during rebuild, including device resources and all retained host metadata. A smaller active-pair capacity is not a bound on total memory.

Current FI1-FI8 behavior remains the refinement oracle: the same admitted observations must reproduce the same movement, energy, repair and persistence history through every allowed storage version.

Psi-index reference and acceptance

FORMAL PENDING | PX8 | LITERAL VECTORS AND OBLIGATIONS

PX8. Default 4 by 5 reference

Use the default Klein recipe (centre 0, radius 2, turns [11,53,137]) and binding epoch 0, sign +1, phase origin 0. The literal orders below follow PX1-PX5. The retained docs/evidence/psi-f8-v1/formal-reference.json contains all 20 node records and the complete tree; these are formal vectors, not runtime measurements.

sorted node IDs:
18,17,19,15,16,4,3,14,13,1,2,11,12,9,0,5,10,6,8,7
preorder node IDs:
1,16,17,18,19,15,3,4,14,13,0,11,2,12,9,6,5,10,8,7

Node ID	Five-component canonical key
0	(3,2,0,0,0)
6	(3,3,1,22,6)
7	(4,3,2,75,7)
16	(1,3,1,22,16)
17	(0,3,2,75,17)

The root is row 0, node 1; its left child is row 1, node 16, and its right child is row 10, node 0. The tree relation 1 to 16 is not a geometric movement edge. Node 16 has $g=(-2,2)$, $\lambda=8$ and $\Psi=(-1,1)$. Its temporary alias (7,-1) has reflected-chart components $g=(-2,-2)$, $\Psi=(-1,-1)$, but the same canonical key.

Required independent conformance

Require independent cover/BFS and field oracles; all 25 possible gradients, both signs and both local frames; eigen-equations, zero-axis tie and bounds; literal keys, complete rows and phase derivations; negative/large aliases; strict binding types, MISS and corrupted trees.

Require real GPU construction and tree-governed texture lookup, corrupted-row rejection and no CPU fallback; actual sample eviction/regeneration; capacity and rebuild invariance including retained DEFER search; complete candidate certification, atomic failure, cleanup-after-commit reporting and epoch overflow. Legacy archives remain bit-exact, and the entire FI8 mission remains unchanged.

Revision 3 acceptance status: FORMAL PENDING. PX1-PX8 are bound before runtime coding. The earlier 368 passing tests, 41 actual-device methods and 23 field-agent checks are preserved evidence for the prior profiles. They are not counted as execution evidence for this new Psi/index layer.

This finite binding supplies local SDF-derived eigenstructure and a canonical middle-out storage index. It does not complete global Psi traversal, Hadamard routing, geometry-changing growth, active scale transitions, physical adapters or indefinite continuation.